Operating System
CS-2006
Lecture 1

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What is Operating System?

- A program that acts as an intermediary between a user of a computer and the computer hardware
- Operating system goals:
  - Execute user programs and make solving user problems easier
  - Make the computer system convenient to use
  - Use the computer hardware in an efficient manner

Structure of Computer System

- Computer system can be divided into four components:

Hardware

Provide basic Computing resources

Examples

- CPU
- Memory
- I/O

Operating System

Controls and coordinates use of hardware among various applications and users

Examples

- Linux
- Windows

User

Who interact with system

Example

- People
- Machine
- Other Computers

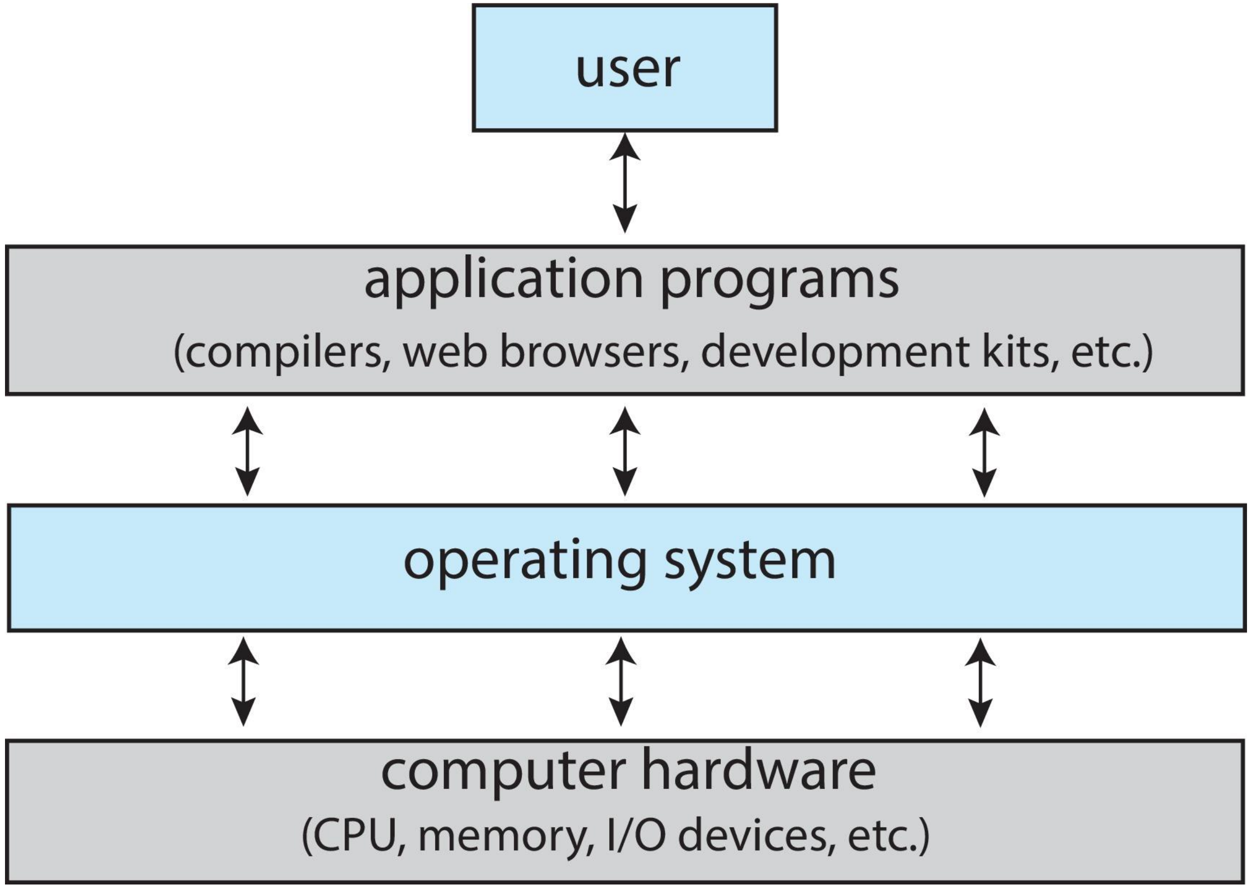
Application Program

define the ways in which the system resources are used to solve the computing problems of the users

Example:

- Word processor
- Compiler
- Database

Abstract View of Computer System



What an Operating System Can do?

In user point of view users want convenience in term of

- Ease to use
- Good performance (not care about resource utilization

In shared Computer such as mainframe or minicomputer

- Important to make all users happy
- OS make efficient use of resource allocator and control program

In dedicated path such as workstation

- Frequently provide the resources from server without any delay

In mobile devices such as smart phones and tablets

- Optimized for usability of battery life
- User Interface Like touch screen

No user interface system Embedded System in device

- Run primarily without user intervention

Basic Operating System Definition/Terms

1. ***Kernel***
   - Program that run all the time on computer
   - Primary Interface between hardware and process

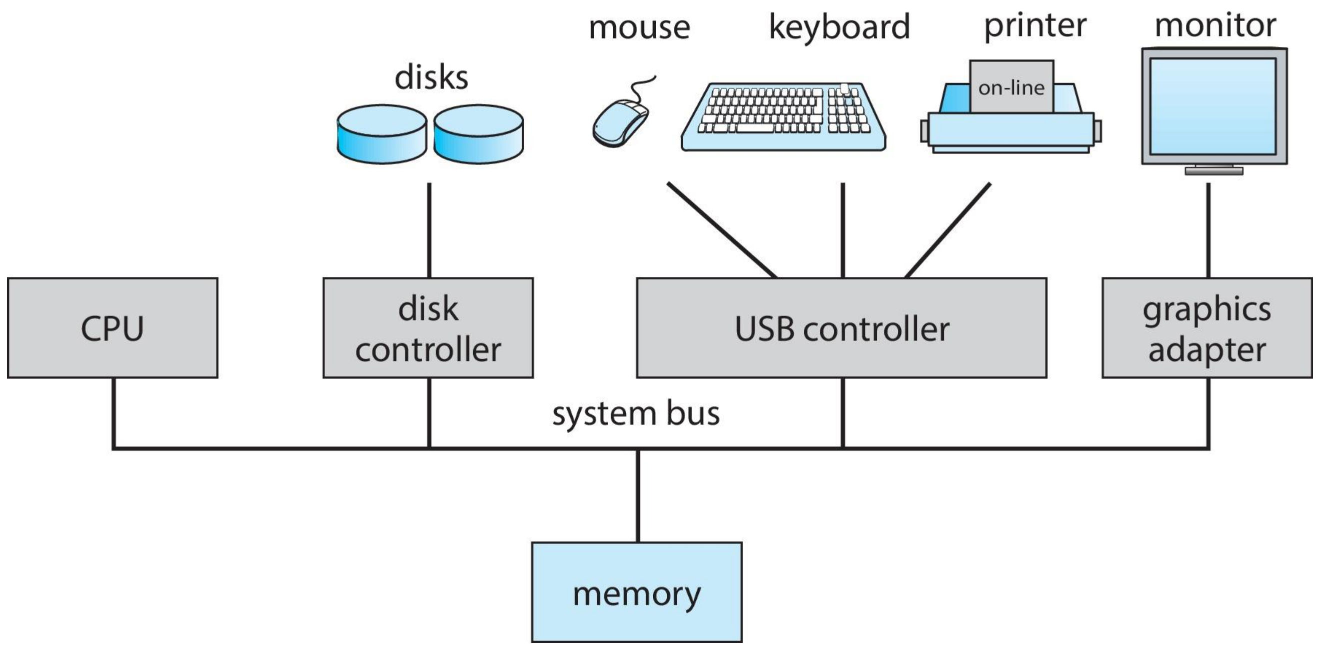
2. ***System Program***
   - Associated with operating system but not part of kernel

3. ***Application Program***
   - all programs not associated with the operating system

4. ***Middleware***
   - Middleware is software that lies between an operating system and the applications running on it.
     - Example: databases, multimedia, graphics

Computer System Operations

- Basic Operations
  - Device Controller
  - Concurrent Execution



device controllers connect through common bus providing access to shared memory

Concurrent execution of CPUs and devices competing for memory cycles

Computer System Operations (Cont…)

- Concurrent Operations
  1. I/O and CPU can execute concurrently
  2. CPU moves data from/to main memory to/from local buffers

- Device Controller
  1. Each device controller is in ***charge of a particular device type***
  2. Each device controller has a ***local buffer***
  3. Each device controller type has ***an operating system device driver*** to manage it

Buffer

The buffer is an area in the main memory used to store or hold the data temporarily.

Interrupts

An interrupt is a signal emitted by hardware or software when a process or an event needs immediate attention

- Operating system is an interrupt driven.
- Basic Terms of Interrupt
  1. Interrupt Vector:
     - Transfer control to interrupt service
     - Contain address of all services

  2. Interrupt architecture
     - Save the address of interrupted instruction

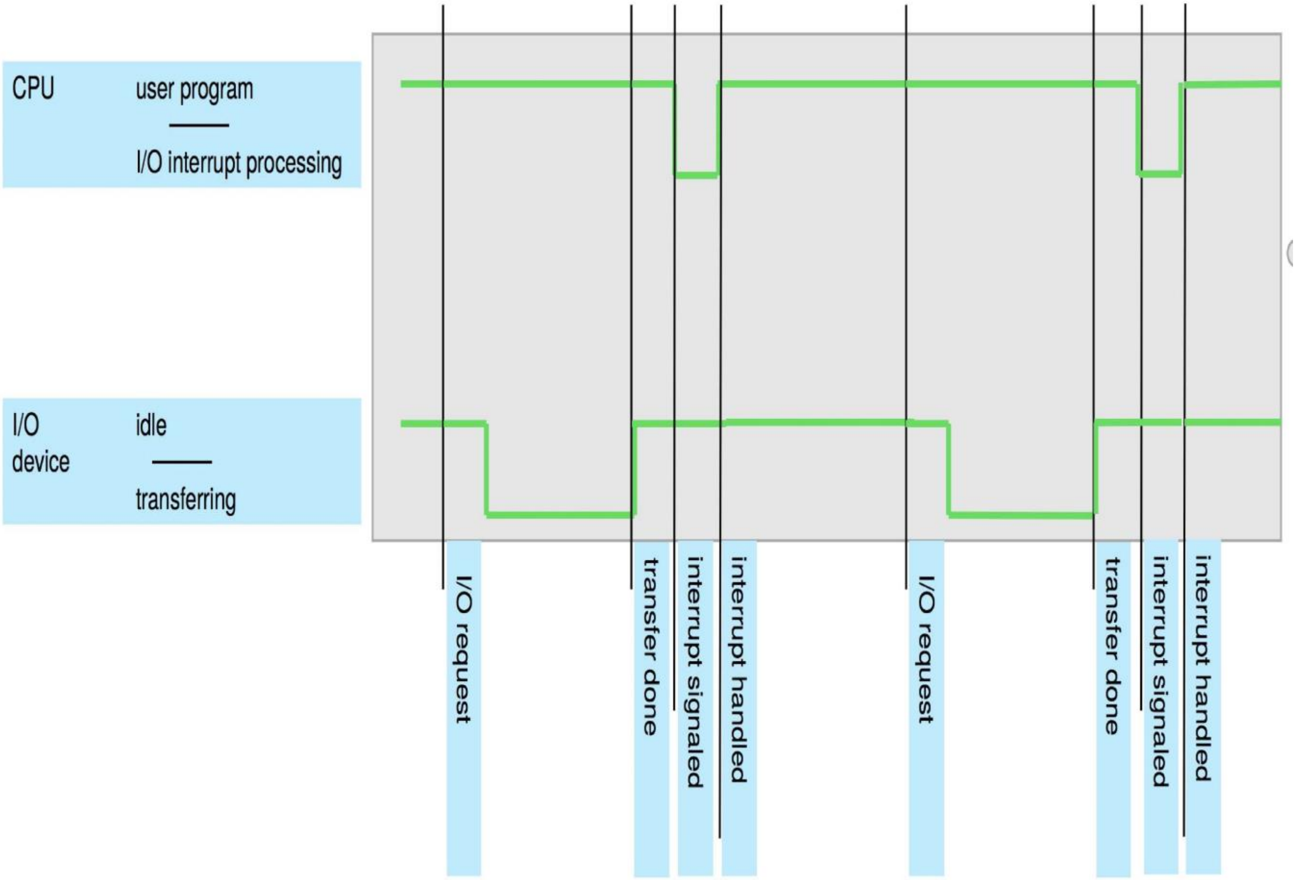
  3. Interrupt trap/execution
     - software-generated interrupt caused either by an error or a user request

Interrupt Timeline

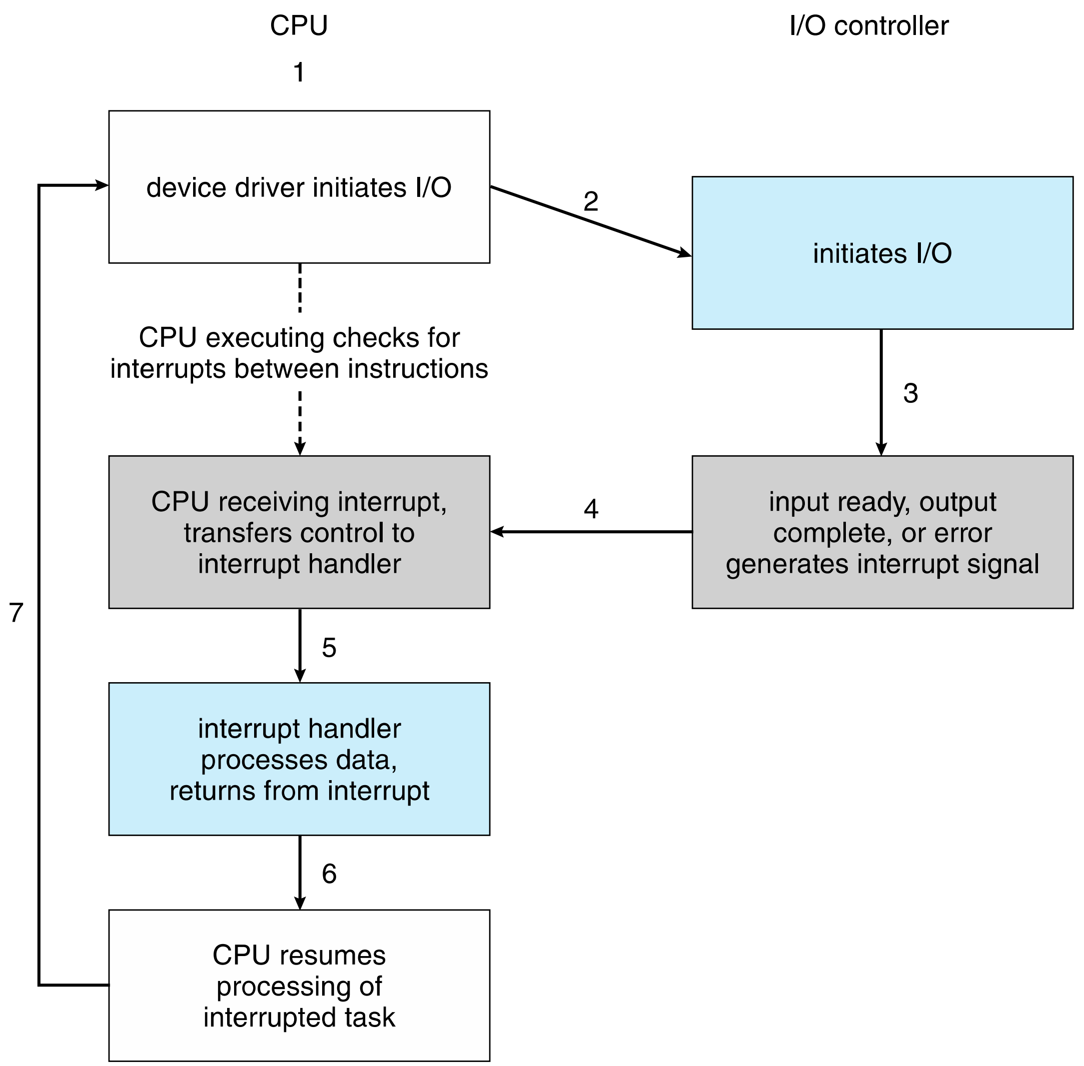
1. The operating system preserves the state of the CPU by storing the registers and the program counter

2. Determines which type of interrupt has occurred:

3. Separate segments of code determine what action should be taken for each type of interrupt



Interrupt Driven I/O cycle



I/O Structure

Method 1: wait for completion

After I/O starts, control returns to user program only upon I/O completion

1. Wait instruction idles the CPU until the next interrupt
2. Wait loop (contention for memory access)
3. At most one I/O request is outstanding at a time, no simultaneous I/O processing

Method 2: without wait for completion

After I/O starts, control returns to user program without waiting for I/O completion

1. **System call** – request to the OS to allow user to wait for I/O completion
2. **Device-status table** contains entry for each I/O device indicating its type, address, and state
3. OS indexes into I/O device table to determine device status and to modify table entry to include interrupt

Bootstrap

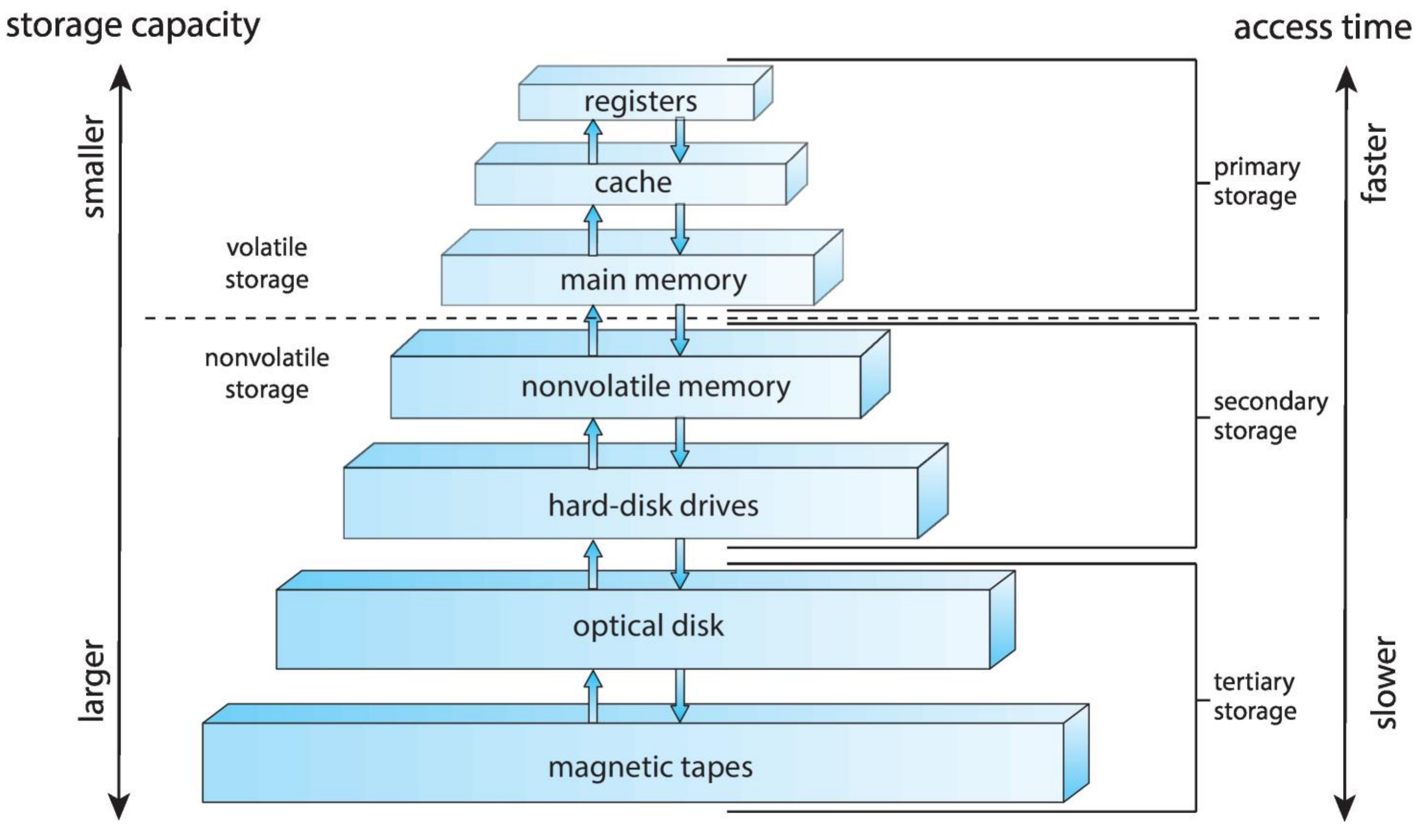
- Bootstrap means to power on the system, which
  - Initialize all basic aspects/software of system
  - Load operating system, kernel and start execution

- Firmware
  - The firmware is embedded in the hardware and may not be changed.
  - Its non-volatile.

Storage Structure

Storage	Main Memory	only large storage media that the CPU can access directly	**Random access** Typically **volatile** Typically, in the form of **Dynamic Random-access Memory (DRAM)**
	Secondary Storage	extension of main memory	provides large **nonvolatile** storage capacity

Storage Structure



Storage Structure (Cont…)

- Registers
  - Fastest type of storage in a computer system, but they are also the smallest.
  - Used to hold data that the CPU needs to access quickly

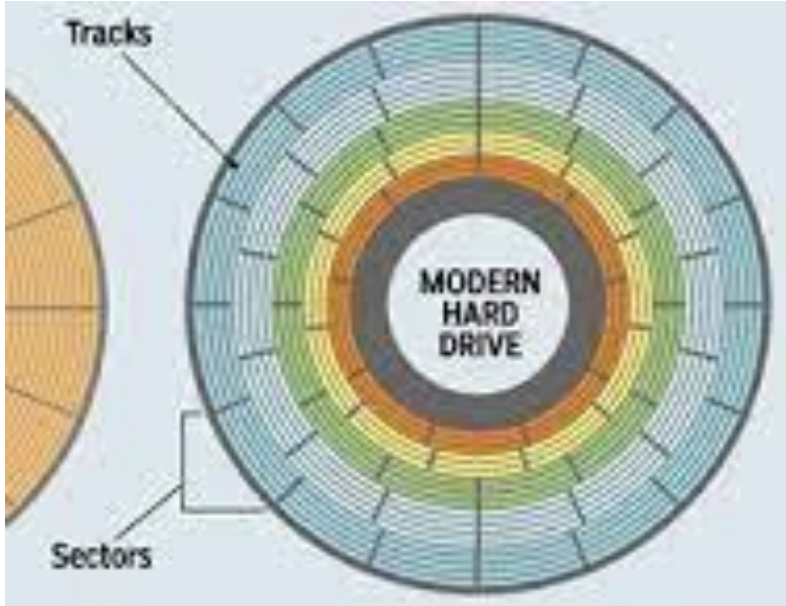
- Cache
  - Small amount of high-speed memory that is used to temporarily store frequently accessed data

- Main Memory
  - primary storage used by a computer system.
  - larger than cache memory, but it is also slower.

- Non-Volatile Memory
  - Type of computer memory that can retain stored information even after power is removed.
  - Becoming more popular as capacity and performance increases, price drops

Storage Structure (Cont…)

- Hard disk drivers: Type of data storage device that is used in laptops and desktop computers
  - Tracks
    - Each platter of a hard disk is divided into a number of tracks.
  - Sectors
    - Each track is divided into a number of sectors, each of which can store the same amount of data. A sector is the smallest physical storage unit on the disk, and on most file systems it is fixed at 512 bytes in size

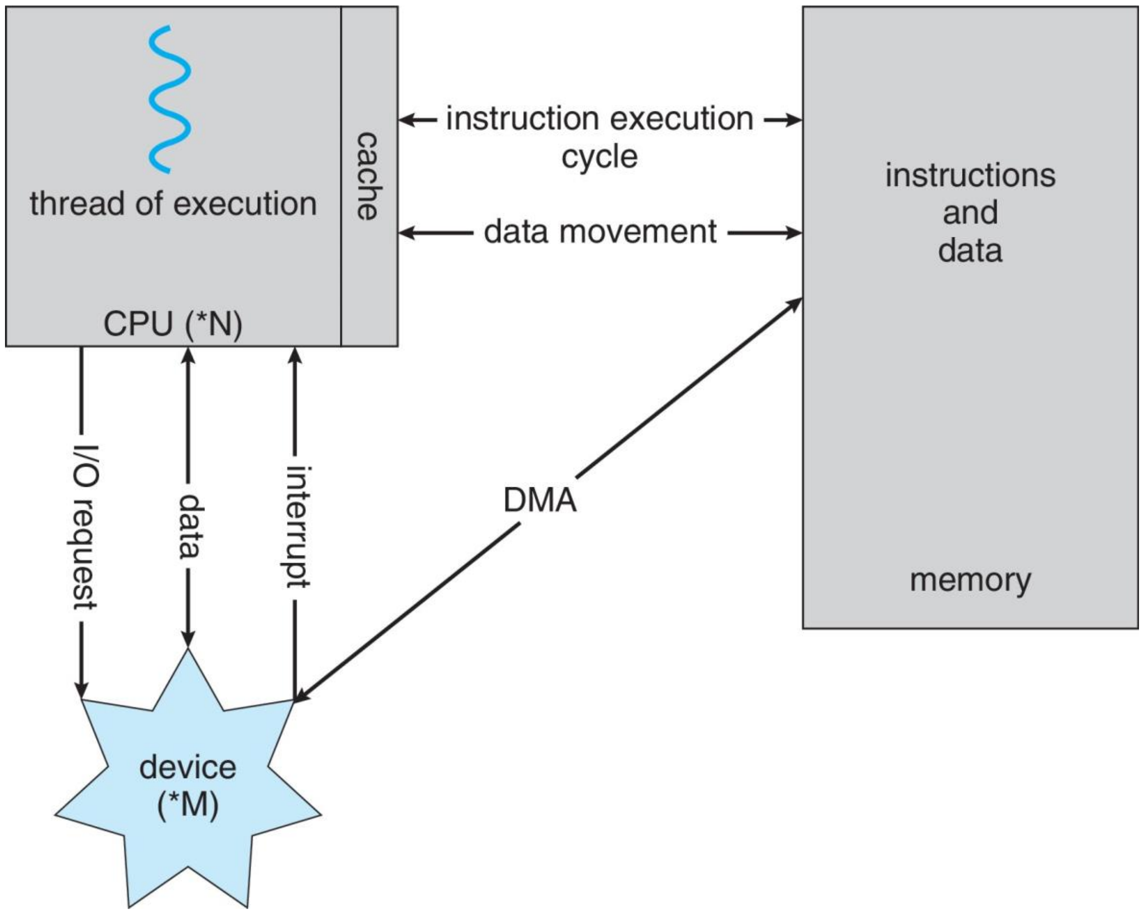


Storage Structure (Cont…)

- Optical Disk
  - An optical disk is an electronic data storage medium that can be written to and read from using a low-powered laser beam
  - CD, DVD, Blu-ray, etc., are the examples of optical disks.

- Magnetic Disk
  - Magnetic disks are flat circular plates of metal or plastic, coated on both sides with iron oxide.
  - Hard Disk, Floppy Disk, Magnetic Tape, etc. are examples of magnetic disks

How Modern Computer Work?



Direct Memory Access Structure (DMA)

- Direct memory access (DMA) is the process of transferring data without the involvement of the processor itself.
  - Only one interrupt is generated per block, rather than the one interrupt per byte

- Device controller transfers blocks of data from buffer storage directly to main memory without CPU intervention

Operating System Operations

- Bootstrap Program
  - simple code to initialize the system, load the kernel
- Kernel Mode
  - some privileged instructions that can only be executed in kernel mode.
  - Example: interrupt instructions, input output management
- System Daemons
  - Daemon is a type of computer program that runs in the background, performing various tasks without direct interaction from the user.
- Interrupt Driven
  - Hardware interrupt by one device
  - Software interrupt are
    - Software error (e.g., division by zero)
    - Request for operating system service – **system call**
    - Other process problems include infinite loop, processes modifying each other or the operating system

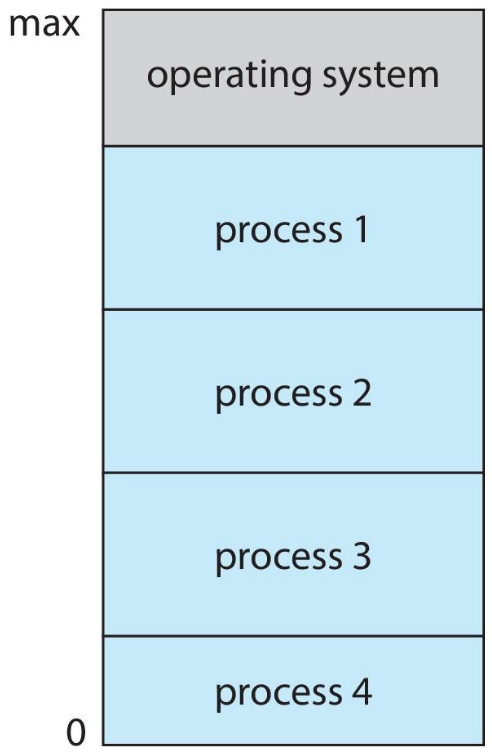
Operating System Operations

- Multi Programming (Batch System)
- Multi Tasking Operations (Time Sharing System)
- Dual Mode operations
- Timer

Multi Programming (Batch System)

- A multiprogramming operating system is one that can execute multiple programs simultaneously on a single processor.
- Single user cannot always keep CPU and I/O devices busy
- One job selected and run via **job scheduling**
- When job has to wait (for I/O for example), OS switches to another job

- Job Scheduling
  - Job scheduling is the process of allocating system resources to many different tasks by an operating system (OS).



Multi Tasking (Time sharing System)

1. Multi-Tasking Operating System is capable of executing multiple application simultaneously without slowing down the system
2. share similar processing resources like a **CPU**.
3. Used CPU scheduling
4. If processes don’t fit in memory, ***swapping moves*** them in and out to run
5. Virtual memory allows execution of processes not completely in memory

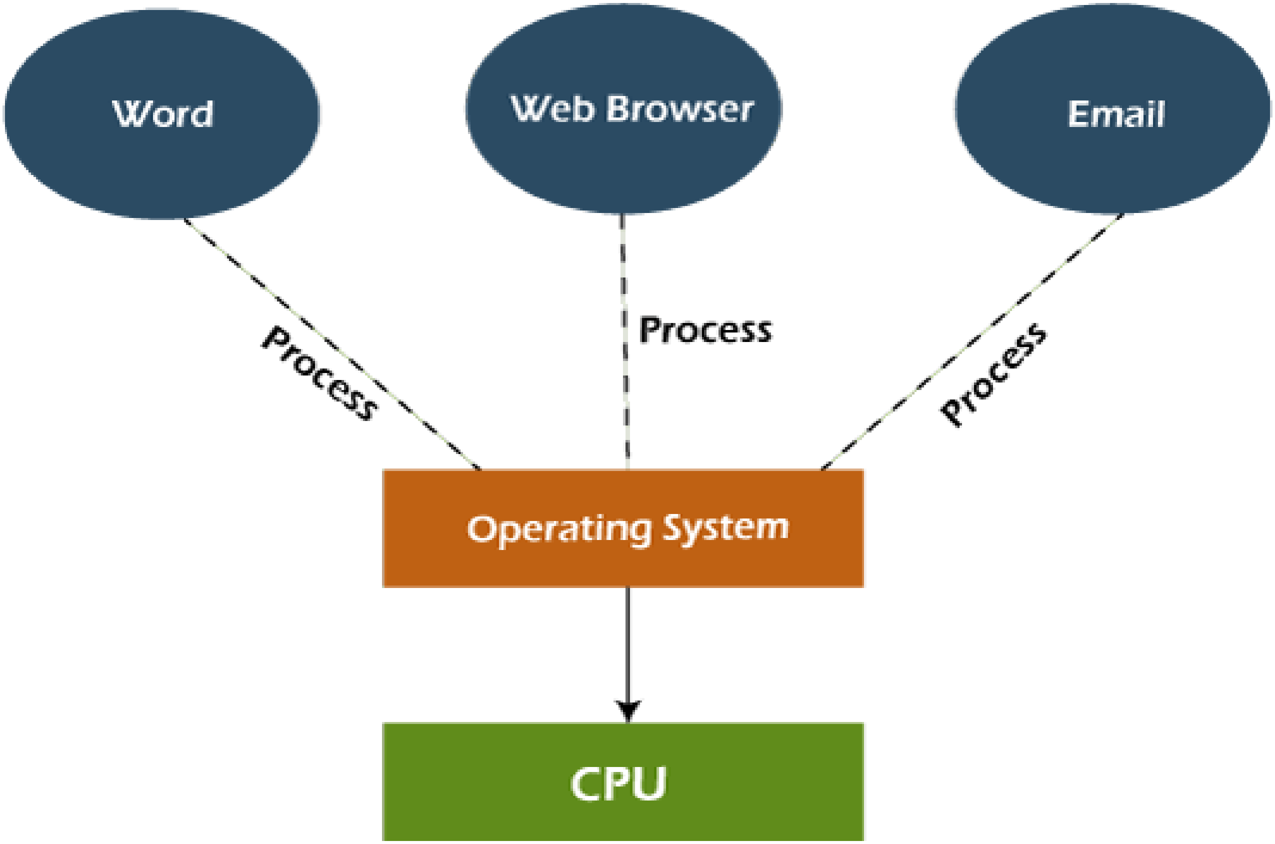
CPU scheduling

CPU Scheduling is a process of determining which process will own CPU for execution while another process is on hold.

Virtual Memory

Virtual memory uses both hardware and software to enable a computer to compensate for physical memory shortages, temporarily transferring data from random access memory (RAM) to disk storage.

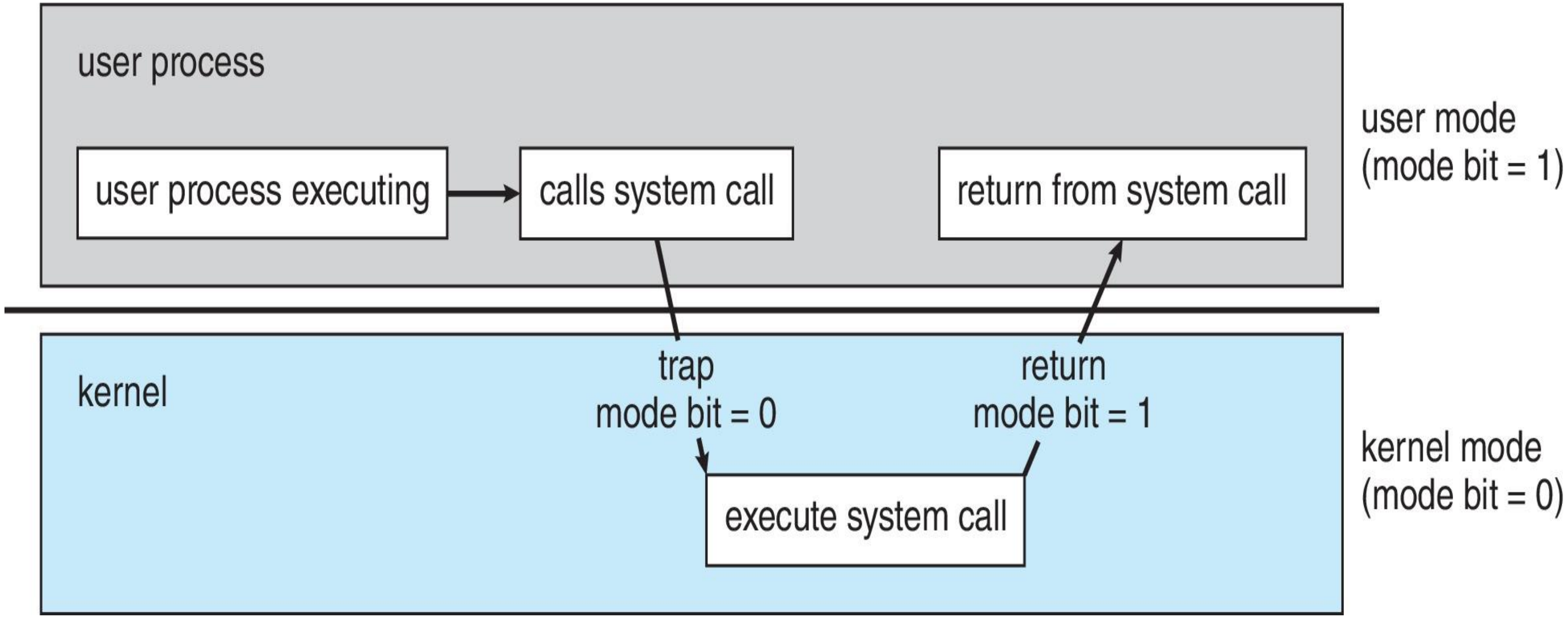
Multi Tasking (Time sharing System) Cont…)



Dual Mode System

- Dual-mode operation is designed to provide a layer of protection and stability to computer systems by separating user programs.
- operating system into two modes
  - **user mode**
    - application program do not have direct access to system resources
  - **kernel mode** (also called supervisor mode, system mode, privileged mode)
    - the processor mode that enables software to have full and unrestricted access to the system and its resources.
  - **Mode Bit**
    - Used to indicate the current mode of the system
      - 0 represent the kernel mode
      - 1 represent the user mode

Dual Mode System (Cont….)



Timer

- Timer to prevent infinite loop (or process hogging resources)
  - Timer is set to interrupt the computer after some time period
  - Keep a counter that is decremented by the physical clock
  - Operating system set the counter (privileged instruction)
  - When counter zero generate an interrupt
  - Set up before scheduling process to regain control or terminate program that exceeds allotted time